SOLUTION OF 1-D WAVE EQUATION

Consider the One-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{——— (1)}$$

Suppose we wish to solve equation (1) for $0<x<a$ and $t>0$ under the boundary conditions

$u(0,t) = \phi(t)$, $u(a,t) = \psi(t)$, $t \geq 0$

where $\phi(t)$ and $\psi(t)$ are specified function of t $(t \geq 0)$ and the initial conditions

$$u(x,0) = f(x), \quad 0<x<a$$

$$\frac{\partial u}{\partial t}(x,0) = g(x), \quad 0<x<a$$

where $f(x)$ and $g(x)$ are specified functions of x (for $0<x<a$)

We know that $\frac{\partial^2 u}{\partial x^2} = \frac{1}{h^2}\left[u_{i+1,j} - 2u_{i,j} + u_{i-1,j}\right]$

$$\frac{\partial^2 u}{\partial t^2} = \frac{1}{k^2}\left[u_{i,j+1} - 2u_{i,j} + u_{i,j-1}\right]$$

Then (1) $\Rightarrow$

$$\frac{c^2}{h^2}\left[u_{i+1,j} - 2u_{i,j} + u_{i-1,j}\right] = \frac{1}{k^2}\left[u_{i,j+1} - 2u_{i,j} + u_{i,j-1}\right]$$

$$\Rightarrow u_{i,j+1} = \frac{c^2k^2}{h^2}\left[u_{i+1,j} + u_{i-1,j}\right] + 2u_{i,j} - \frac{2c^2k^2}{h^2}u_{i,j} - u_{i,j-1}$$

let ~~$\alpha = \frac{k}{h}$~~ $\beta^2 = \frac{c^2 k^2}{h^2}$

$$u_{i,j+1} = 2(1-\beta^2)u_{i,j} + \beta^2(u_{i-1,j} + u_{i+1,j}) - u_{i,j-1}$$

Note :- The method is stable when $\beta \leq 1$ and unstable when $\beta > 1$

$$\beta = \frac{ck}{h} \qquad \beta = 1 \Rightarrow k = \frac{h}{c}$$

$\beta = 1$,

$$\boxed{u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1}}$$

1. By using the explicit three-level formula, solve the wave equation $\frac{\partial^2 u}{\partial t^2} = 4\frac{\partial^2 u}{\partial x^2}$ subject to the boundary conditions

$$u(0,t) = u(4,t) = 0, \quad t \geq 0$$

and the initial conditions

$$u(x,0) = x(4-x), \quad \frac{\partial u}{\partial t}(x,0) = 0, \quad 0 < x < 4$$

Take $h=1$ and $k = 1/2$ and compute $u(x,t)$ for $0 < t \leq 2$.

Solution :- Given $h=1$, $k=1/2$, $c^2 = 4$, $\beta = \frac{c^2k^2}{h^2} = 1$

Given initial conditions

$$u(x,0) = x(4-x)$$

$$u(x_i,0) = x_i(4-x_i)$$

$$u_{i,0} = ih(4-ih)$$

$$u_{i,0} = i(4-i)$$

$$\Rightarrow u_{1,0} = 3, \quad u_{2,0} = 4, \quad u_{3,0} = 3$$

$$u(0,t) = 0$$

$$\Rightarrow u_{0,0} = 0, \quad u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = 0$$

$$u(4,t) = 0$$

$$u_{4,0} = u_{4,1} = u_{4,2} = u_{4,3} = u_{4,4} = 0$$

By the three level explicit formula, we have

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \quad \text{——(1)}$$

put $j=0$

$$u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,-1}$$

$$2u_{i,1} = u_{i+1,0} + u_{i-1,0}$$

$$\boxed{u_{i,1} = \frac{1}{2}\left[u_{i+1,0} + u_{i-1,0}\right]}$$

Given $\frac{\partial u}{\partial t}(x,0) = 0$.

$$\Rightarrow \frac{u_{i,j+1} - u_{i,j-1}}{2k} = 0.$$

$$\Rightarrow u_{i,j+1} = u_{i,j-1}$$

when $j=0$.

$$u_{i,1} = u_{i,-1}$$

Second time level at $t=1$

$$u_{i,2} = u_{i+1,1} + u_{i-1,1} - u_{i,0}$$

j \ i	0	1	2	3	4
0	0	3	4	3	0
1	0	$\frac{4+0}{2}=2$	$\frac{3+3}{2}=3$	$\frac{4+0}{2}=2$	0
2	0	0+3-3=0	2+2-4=0	3+0-3=0	0
3	0	0+0-2	0+0-3	0+0-2	0
4	0	0-3+0	-2-2-0	-3-0+0	0

j \ i	0	1	2	3	4
0	0	3	4	3	0
1	0	2	3	2	0
2	0	0	0	0	0
3	0	-2	-3	-2	0
4	0	-3	-4	-3	0

(2) Evaluate the pivot values of the equation $u_{tt} = 16\, u_{xx}$ taking $h=1$ upto $t=1$. The boundary conditions are $u(0,t) = u(5,t) = 0$, $u_t(x,0) = 0$ and $u(x,0) = x^2(5-x)$

Solution :- $c^2 = 16$, $h=1$, $\frac{k^2 c^2}{h^2} = \beta$

$$k^2 = \frac{h^2}{c^2} \Rightarrow k^2 = 1/16 \Rightarrow k = 1/4$$

∴ We have

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1}$$

Now $u(0,t) = 0 \Rightarrow \boxed{u_{0,j} = 0}$

$$u(5,t) = 0 \Rightarrow \boxed{u_{5,j} = 0}$$

$$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = 0$$

$$u_{5,0} = u_{5,1} = u_{5,2} = u_{5,3} = u_{5,4} = 0$$

$u(x,0) = x^2(5-x)$

$u(x_i,0) = x_i^2(5-x_i)$ $\qquad x_i = ih$

$u_{i,0} = i^2(5-i)$ $\qquad h=1,\ x_i = i$

$u_{1,0} = 4,\quad u_{2,0} = 12,\quad u_{3,0} = 18,\quad u_{4,0} = 16$

Initial Condition $u_t(x,0) = 0$

$$\Rightarrow \frac{u_{i,j+1} - u_{i,j-1}}{2k} = 0 \quad \text{when } j = 0$$

$$\Rightarrow u_{i,1} = u_{i,-1}$$

j \ i	0	1	2	3	4	5
0	0	4	12	18	16	0
1	0	$=\frac{1}{2}(0+12)$ 6	$=\frac{1}{2}(4+18)$ 11	$=\frac{1}{2}(12+16)$ 14	$=\frac{1}{2}(18+0)$ 9	0
2	0	$=0+11-4$ 7	$6+14-12$ 8	$11+9-18$ 2	$14+0-16$ -2	0
3	0	$=0+8-6$ 2	$7+2-11$ -2	$8-2-14$ -8	$2+0-9$ -7	0
4	0	$=0-2-7$ -9	$2-8-8$ -14	$-2-7-2$ -11	$-8+0+2$ -6	0

(3) solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ with boundary conditions

$u(0,t) = 0 = u(1,t)$; $t \geq 0$ and initial conditions

$u(x,0) = \sin \pi x$, $\frac{\partial u}{\partial t}(x,0) = 0$, $0 < x < 1$

taking $h = k = 0.2$ at $t = 1$

Solution :- We have $c^2 = 1$, $c = 1$ $\quad k = \frac{h}{c}$

We $u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1}$

$\frac{\partial u}{\partial t}(x,0) = 0 \Rightarrow u_{i,1} = \frac{1}{2}\left[u_{i-1,0} + u_{i+1,0}\right]$

$u(0,t) = 0 \Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = u_{0,5} = 0$

$u(1,t) = 0 \Rightarrow u_{5,0} = u_{5,1} = u_{5,2} = u_{5,3} = u_{5,4} = u_{5,5} = 0$

$u(x,0) = \sin \pi x$

$\Rightarrow u_{1,0} = \sin \frac{\pi}{5} = 0.5878$

$u_{2,0} = \sin \frac{2\pi}{5} = 0.9511$

$u_{3,0} = \sin \frac{3\pi}{5} = 0.9511$

$u_{4,0} = \sin \frac{4\pi}{5} = 0.5878$

j \ i	0	1	2	3	4	5
0	0	0.5878	0.9511	0.9511	0.5878	0
1	0	$=\frac{1}{2}(0+0.9511)$ 0.4756	$=\frac{1}{2}(0.5878+0.9511)$ 0.7695	$=\frac{1}{2}(0.9511+0.5878)$ 0.7695	$=\frac{1}{2}(0.9511+0)$ 0.4756	0
2	0	$=0+0.7695-0.5878$ 0.1817	$=0.4756+0.7695-0.9511$ 0.294	$0.7695+0.4756-0.9511$ 0.294	$0.7695+0-0.5878$ 0.1817	0
3	0	$=0+0.294-0.4756$ -0.1816	$=0.1817+0.294-0.7695$ -0.2938	$0.294+0.1817-0.7695$ -0.2938	$0.294+0-0.4756$ -0.1816	0
4	0	-0.4755	-0.7694	-0.7694	-0.4755	0
5	0	-0.5878	-0.9511	-0.9511	-0.5878	0

(4) By using the explicit three level formula, solve the wave equation $4\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ subject to the following conditions

$$u(0,t) = u(1,t) = 0 \qquad t \geq 0$$

$$u(x,0) = \begin{cases} 5x/3 & 0 < x \leq 3/5 \\ \frac{5(1-x)}{2} & \frac{3}{5} < x < 1 \end{cases}$$

$$\frac{\partial u}{\partial t}(x,0) = 0$$

Taking $h = 1/5$ Carry out Computations upto $t = 2/5$.

Solution :- $h = 1/5 \quad c = 2 \Rightarrow k = \frac{h}{c} = \frac{1}{10}$

Here $x_0 = 0$ and $x_5 = 1$

$u(0,t) = 0 \Rightarrow u_{0,j} = 0$

$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = 0$

$u(1,t) = 0 \Rightarrow u_{5j} = 0$

$\Rightarrow u_{5,0} = u_{5,1} = u_{5,2} = u_{5,3} = u_{5,4} = 0$

Given $u(x,0) = \begin{cases} \frac{5x}{3} & 0 < x \le 3/5 \\ \frac{5(1-x)}{2} & \frac{3}{5} < x < 1 \end{cases}$

$x = ih$
$= i/5$

$u_{i,0} = \begin{cases} \frac{5ih}{3} & i = 1, 2, 3 \\ \frac{5(1-ih)}{2} & i = 4 \end{cases}$

$u_{i,0} = \begin{cases} i/3 & i = 1, 2, 3 \\ \frac{5(1-i/5)}{2} & i = 4 \end{cases}$

$u_{1,0} = 1/3, \quad u_{2,0} = 2/3, \quad u_{3,0} = 1$

$u_{4,0} = \frac{5}{2}\left(1 - \frac{4}{5}\right) = 1/2$

Also $\frac{\partial u}{\partial t}(x,0) = 0 \Rightarrow u_{i,1} = u_{i,-1}$

By three level explicit formula

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \quad \text{——— (1)}$$

For the first level, put $j = 0$ in (1)

$$\Rightarrow u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,-1}$$

$$\Rightarrow u_{i,1} = \frac{1}{2}\left[u_{i+1,0} + u_{i-1,0}\right]$$

j \ i	0	1	2	3	4	5
0	0	1/3	2/3	1	1/2	0
1	0	=(0+2/3)/2 = 1/3	=1/2(1/3+1) 2/3	7/12	1/2	0
2	0	1/3	1/4	1/6	1/12	0
3	0	1/12	-1/6	-1/4	-1/3	0
4	0	-1/2	-5/12	-2/3	-1/3	0